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Measurement and modeling of BRDF for stray light suppression in gravitational wave telescope baffle surfaces

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ABSTRACT

Space gravitational wave detection requires highly sensitive capture of extremely weak gravitational wave signals, posing unprecedented challenges to the stability and noise suppression capabilities of telescope optical systems. As a critical component at the optical input of gravitational wave telescope, the surface properties of the baffle directly affect the distribution of stray light and the system's sensitivity to weak signals. Based on a self-developed Bidirectional Reflectance Distribution Function (BRDF) measurement system, this paper systematically measured the surface scattering characteristics of three typical baffle materials in the visible and near-infrared bands, and deeply analyzed the influence of surface morphology, incident angle and wavelength on BRDF characteristics. In addition, based on the measured data, a BRDF mathematical model accurately characterizing the scattering behavior of materials was established, and its parameters were optimized using the simulated annealing algorithm. The research results not only provide essential data support for stray light ray-tracing calculations in baffle design but also offer important theoretical guidance for optimizing the optical systems and stray light suppression strategies of gravitational wave telescope.

1. Introduction

The first direct detection of gravitational waves opened a new era in scientific exploration, providing an unprecedented means to study extreme astrophysical phenomena in the universe [1,2]. Compared to ground-based gravitational wave detectors, space-based observatories offer significant advantages, including sensitivity to lower frequency bands and the ability to detect more distant astrophysical sources, making them a crucial avenue for future gravitational wave astronomy [3–5]. However, due to the extremely weak nature of gravitational wave signals, space-based detection systems demand exceptionally high precision in optical measurements and stringent control of stray light [6–8]. In space gravitational wave detection missions, such as China's TianQin and Taiji programs and the Laser Interferometer Space Antenna (LISA) led by the European Space Agency (ESA), telescopes must simultaneously perform 1064 nm laser emission and high-precision reception of extremely weak echo signals [9,10]. Since laser beams experience severe attenuation during long-distance propagation and must undergo multiple reflections before reaching the detector, background light from celestial bodies and scattered light from the surfaces of opto-mechanical components may enter the system as stray light. This introduces additional optical path differences and phase noise, thereby reducing the accuracy of interferometric measurements and degrading the signal-to-noise ratio (SNR). In extreme cases, stray light may even annihilate the target gravitational wave signal, resulting in mission failure [11–13]. As a critical optical component in space gravitational wave telescopes, the baffle not only acts as a physical shield to minimize stray light intrusion but also precisely directs the optical path, effectively mitigating non-target light interference and ensuring a high SNR [14,15]. The design and performance of the baffle are directly related to the system's ability to detect weak gravitational wave signals, making it one of the core elements in stray light suppression [16,17].

In recent years, multiple research teams have conducted in-depth studies on stray light suppression and baffle design for gravitational

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wave telescopes. For example, Canuel et al. [18] investigated the issue of backscattering in input optical systems and utilized the Bidirectional Reflectance Distribution Function (BRDF) to calculate stray light recoupling noise, providing a theoretical basis for stray light suppression in gravitational wave detectors. Zeidler et al. [19] developed a hemispherical BRDF measurement device to evaluate the scattering characteristics of Ni-P coatings, metals, and carbon-based coatings at 1064 nm, quantifying the potential applications of different materials in gravitational wave telescopes. Wang et al. [20] analyzed the impact of backscattered light in space-based gravitational wave telescopes and proposed suppression strategies based on optical axis deflection and optical surface optimization, offering important references for stray light suppression and baffle design. DeSalvo et al. [21] proposed a helical baffle design for gravitational wave detection, which enhances stray light suppression by absorbing scattered light through multiple reflections. Akutsu et al. [22] conducted simulation analyses on the stray light issues in the central interferometric region of the KAGRA gravitational wave telescope and optimized the extinction coatings and geometric structure of the baffles to reduce parasitic scattering from material surfaces. Andrés-Carcasona et al. [23] studied the stray light problem in the main arms of the Einstein gravitational wave telescope and proposed a baffle optimization scheme based on surface roughness control, effectively reducing scattered light recoupling while improving the optical performance of the system.

In the optical system of gravitational wave telescopes, the optical reflection properties of the baffle material surface are a key factor in determining the distribution of stray light, and their quantitative characterization is crucial for system design. As a fundamental method for quantifying surface reflection characteristics [24], BRDF measurement provides the theoretical foundation and data support for characterizing the scattering behavior of baffle materials under multi-angle and multi-wavelength conditions. Meanwhile, BRDF modeling also plays a significant role in surface analysis and optical system design [25]. Mainstream models such as the Cook-Torrance [26], Torrance-Sparrow [27], and Davies models focus on specular reflection on rough surfaces, microfacet-based mirror reflection theory, and multi-component reflection descriptions, respectively [28]. Wu et al. [29] extended the Torrance-Sparrow model and proposed a five-parameter statistical model to accommodate non-Gaussian distributions; Ďurikovič and Mihalik [30] modeled BRDF based on the Cook-Torrance microfacet model through spectral reflectance measurements, providing a novel approach for simulating the characteristics of metal surfaces; Wang et al. [31] measured the spectral BRDF characteristics of Acktar aerospace black paint and developed a six-parameter BRDF model, providing technical support for stray light suppression in space optical systems.

Nevertheless, research on the BRDF of gravitational wave telescope baffle materials remains limited, particularly regarding the measurement, analysis, and comprehensive surface property database. Furthermore, the relationship between the topological structure of baffle materials and their BRDF has received minimal attention. To address these gaps, this paper systematically investigates the BRDF measurement and modeling of baffle material surfaces for gravitational wave telescope. A high-precision measurement platform was developed to obtain BRDF data under multi-angle and multi-wavelength conditions. Based on experimental results, a mathematical model was constructed to reflect the true optical behavior of the materials. This study aims to provide a reliable basis for the design of baffles for gravitational wave telescope and to offer scientific support for enhancing the optical performance and system sensitivity of space gravitational wave detection missions.

2. BRDF experimental measurement

2.1. Experiment device and method

BRDF describes the relationship between the incident irradiance and the spatial distribution of scattered radiation modulated by random spatial frequency components of the optical surface, which typically originate from surface roughness introduced during the manufacturing process or from particulate contamination [32]. By measuring and analyzing the BRDF of sample surfaces, a thorough understanding of the material surface's scattering characteristics can be achieved. It is a five-dimensional function about the incident zenith angle θ_i, reflection zenith angle θ_r, incident azimuth angle φ_i, reflection azimuth angle φ_r and wavelength λ. BRDF is defined as the ratio of the outgoing radiance to the incident irradiance, which can be expressed as:

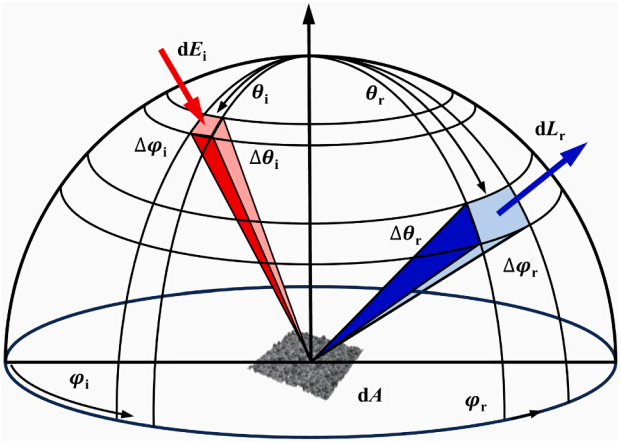


Fig. 1. Geometry and notations in BRDF definition.

$$BRDF(\theta_i,\varphi_i,\theta_r,\varphi_r,\lambda)=\frac{dL_r(\theta_i,\varphi_i,\theta_r,\varphi_r,\lambda)}{dE_i(\theta_i,\varphi_i,\lambda)}=\frac{dL_r(\theta_i,\varphi_i,\theta_r,\varphi_r,\lambda)}{L_i(\theta_i,\varphi_i)\cos\theta_i d\omega_i}\left(\mathrm{sr}^{-1}\right) \tag{1}$$

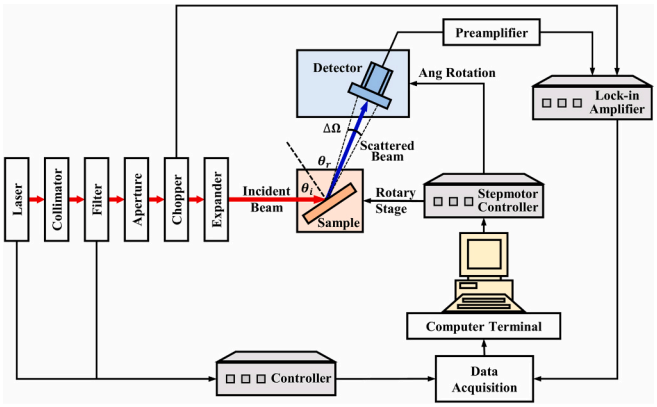
where λ is the wavelength, θ is the zenith angle measured from normal, φ is the azimuth angle, $dE_i(\theta_i,\varphi_i,\varphi_r,\lambda)$ is the incident spectral irradiance measured in unit (W/m^2), and $dL_r(\theta_i,\theta_r,\varphi_i,\varphi_r,\lambda)$ is the reflected spectral radiance and has the unit ($W/(m^2 \bullet sr)$). Each physical quantity is shown in Fig. 1.

The BRDF is a specific form of the more general Bidirectional Scattering Distribution Function (BSDF), which includes both BRDF and Bidirectional Transmittance Distribution Function (BTDF) components. For opaque materials, where no transmission occurs, the BSDF simplifies to the BRDF. One common experimental approach for measuring the BRDF is the Angle Resolved Scattering (ARS) method. ARS is a non-imaging technique [33], which characterizes the angular scattering behavior of a material surface by measuring the scattered optical power ΔP_r collected by the detector within a solid angle $\Delta\Omega_r$ around the reflection direction (θ_r,φ_r), normalized by the total incident optical power P_i illuminating the sample surface. Considering the geometric projection relationship between radiance and irradiance, the BRDF can be expressed as:

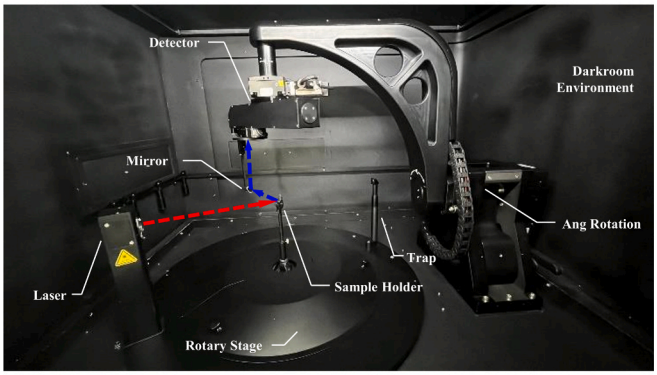
$$BRDF(\theta_i,\varphi_i,\theta_r,\varphi_r,\lambda)=\frac{ARS(\theta_r,\varphi_r,\lambda)}{\cos\theta_r}=\frac{\Delta P_r(\theta_r,\varphi_r,\lambda)}{P_i\Delta\Omega_r\cos\theta_r}\left(\mathrm{sr}^{-1}\right) \tag{2}$$

2.2. Experiment system introduction

Currently, BRDF measurement instruments have been developed and applied worldwide. Representative systems, such as the SOC-210BDR [34] developed by Surface Optics Corporation (USA), the TSW-CASI [35] developed by Scatter Works, Inc. (USA), and the ALBATROSS [36] system developed by the Fraunhofer Institute (Germany), have been widely used for characterizing the scattering properties of optical coatings and engineering materials. Drawing on the design concepts of these mature instruments and addressing the practical measurement requirements of baffle materials with high diffuse reflectance and low



(a) Schematic of the experimental arrangement.



(b) Photograph of the experimental set-up.

Fig. 2. BRDF measurement apparatus scheme and testing photos.

reflectance for space gravitational wave detection missions, a dedicated BRDF measurement system was developed in this work. As shown in Fig. 2, this BRDF measurement device mainly consists of three components: the laser light source system, the detection system, and the mechanical scanning system. In the design of laser source systems, considering the practical application of gravitational wave telescope shields, sunlight, multiple reflections from the internal structural surfaces, and auxiliary beams used for system calibration may introduce light into the shields over a wider spectral range. Therefore, in addition to the nominal operating wavelength of 1064 nm for space gravitational wave detection, the system also covers representative visible wavelengths of 520 nm, 640 nm, and 808 nm. The light source is provided by four single-wavelength solid-state lasers, each with an output power of 100 mW.

After the laser emission, it passes through a defocus system consisting of two off-axis parabolic mirrors with focal lengths of$f_1 = 50.8$ mm and$f_2 = 101.6$ mm to achieve collimation and beam expansion. Then, the beam passes through a filter to select the energy of the light beam, and then through off-axis parabolic mirror with focal length$f_3 = 76.2$ mm and an ellipsoidal mirror with focal length$f_4 = 100$ mm to bend and converge the beam, finally reaching the surface of the test sample, with a spot diameter of approximately 2 mm. To improve collimation and the accuracy of the incident angle, and to suppress some stray light and reduce measurement errors, a pinhole aperture is set at the confocal plane of mirror 3 and mirror 4. Meanwhile, a signal averaging integration strategy was employed during measurement to suppress random speckle noise and improve accuracy. It is worth noting that the aforementioned beam expansion, collimation, and optical path folding components are all integrated into the base of the laser column, making the system more compact while reducing optical axis drift caused by relative position changes of optical elements.

The incident laser, modulated by the test sample surface, forms scattered light in the hemisphere space, which is then received by the detection system after reflection by a plane mirror. The detection system uses two types of detector devices: a photomultiplier tube (R7206-01, Hamamatsu Photonics K.K., JP) in the visible light range and an InGaAs photodiode (G12180-030A, Hamamatsu Photonics K.K., JP) in the near-infrared range. The light signals are converted into corresponding photocurrents by the detectors. After amplification by the corresponding preamplifiers and filtering by common-mode and differential-mode filters, they are converted into voltage signals and outputted to a lock-in amplifier (MFLI500k, Zurich Instruments, CH) through differential operation. The lock-in amplifier simultaneously receives a frequency reference signal from a chopper and a synchronous signal from the detector. The detection data is transmitted to the data acquisition and processing terminal via RS485. The mechanical scanning system employs three electric motorized rotary stages (M−060, Physik Instrumente, DE) to control and achieve fully automatic measurement of BRDF in the 2 π hemisphere space under different incident angles, scattering angles, and scattering azimuth angles. Compared to traditional manual or stepwise scanning methods, the fully automated scanning mechanism of this system enables higher measurement efficiency and repeatability, reduces human operation errors, and improves angular accuracy. To minimize background light scattering in the test environment, the entire measurement is conducted in a dark box environment. Since the solid angle of reflected light received by the detector directly affects the accuracy and resolution of BRDF measurements, the detector used in this study has an effective detection area A of 9 mm^2 and is placed at a distance R of 500 mm from the measured surface. Therefore, the receiving solid angle Ω_r can be calculated as: $\Omega_r = A/R^2 = 9/500^2 = 3.6 \times 10^{-5}$ sr.

During the experimental testing process, the bottom rotating platform of the laser is first adjusted to the initial angle of 0° to facilitate alignment of the sample. The range of angles between the normal to the test sample surface and the incident laser is from −90° to 90°, with measurements taken in groups every 10°. After determining the incident angle, the rotating platform of the detector device is adjusted from −90° to 90°, with data collected at 1° intervals. The system employs a high-precision angular encoder for automatic calibration to minimize potential angular errors introduced during the mechanical scanning process, ensuring the stability and consistency of measurement results at different angles.

2.3. Experimental uncertainty

The accuracy of BRDF measurements is influenced by multiple factors, including the laser light source, detection system, mechanical scanning mechanism, test background environment, solid angle, value transfer and measurement repeatability. Since air scattering in the test environment and human operational errors are negligible, they are not considered in this analysis.

The overall measurement uncertainty is categorized into Type A (statistical) and Type B (systematic) components [37]. The total combined standard uncertainty is calculated in two steps using the root-sum-square (RSS) method. First, the Type A and Type B components are evaluated separately, and then they are combined to obtain the total standard uncertainty, which is given by:

$$U(BRDF)/BRDF = \sqrt{(U_A)^2 + (U_B)^2} \tag{3}$$

Type A uncertainty originates from measurement repeatability. Under identical experimental conditions, each scattering angle was measured independently five times, and statistical analysis yielded a relative standard deviation of 0.86%, denoted as $U(T)/T$. Type B uncertainty consists of six primary contributors, and the combined standard uncertainty is expressed as:

Table 1
Summary of identified uncertainty components of current apparatus.

Source	Uncertainty	Uncertainty value
Laser light source	$U(I)/I$	0.32%
Detection module	$U(M)/M$	0.29%
Rotary scanning device	$U(D)/D$	1.63%
Darkroom environment	$U(C)/C$	0.89%
Solid angle	$U(\Omega_r)/\Omega_r$	0.50%
Value transfer	$U(S)/S$	0.35%
Measurement repeatability	$U(T)/T$	0.86%
BRDF	$U(BRDF)/BRDF$	2.18%

$$U_B = \sqrt{\frac{U^2(I)}{I^2} + \frac{U^2(M)}{M^2} + \frac{U^2(D)}{D^2} + \frac{U^2(C)}{C^2} + \frac{U^2(\Omega_r)}{\Omega_r^2} + \frac{U^2(S)}{S^2}} \quad (4)$$

In this equation, $U(I)/I$, $U(M)/M$, $U(D)/D$, $U(C)/C$, $U(\Omega_r)/\Omega_r$ and $U(S)/S$ represent the uncertainties caused by the laser source system, detection system, mechanical scanning mechanism, darkroom environment, measurement solid angle and value transfer, respectively. The uncertainty of the laser source primarily due to output power fluctuations, with a measured short-term stability of 0.32 % over a two-hour period. The detection system uncertainty arises from the detector's intrinsic noise and gain drift, and through multiple reference sample measurements, its relative uncertainty is determined to be 0.29 %. The uncertainty of the mechanical scanning mechanism primarily stems from angular errors in the rotary platform and repeatability errors in the positioning system. After multiple calibrations with a high-precision angular encoder, its impact on the measurement is quantified as 1.63 %. The uncertainty of the solid angle Ω_r is determined by the detector's effective receiving area A and its distance R from the sample. Considering collimation and alignment errors, its uncertainty is estimated at 0.5%. The environmental noise floor was calibrated before the test. The actual measured radiation flux of the scattered light from the inner wall entering the detector was 1.28×10^{-14} W, while the signal light radiation flux was 1.43×10^{-12} W. The scattered light from the inner wall is two orders of magnitude lower than the signal light, and the uncertainty caused by the inner wall of the darkroom can be calculated to be 0.89 %. This study employed a relative BRDF measurement method using a Spectralon® Lambertian standard from Labsphere, traceable to the National Institute of Standards and Technology (NIST). The reflectance uncertainty is 0.35 % according to the official calibration data. Each parameter is shown in Table 1. Based on this, the overall uncertainty of the system can be calculated to be 2.18 %. With a coverage factor$k = 2$ (corresponding to a confidence level of approximately 95 %), the expanded uncertainty is calculated to be 4.36 %.

To validate our measurement system, the BRDF of a Spectralon® SRM-99 standard Lambertian reflector was measured at four wavelengths. The measurement results are shown in Fig. 3. For an ideal Lambertian reflector, the BRDF is independent of both the incident and scattering angles and depends solely on the diffuse reflectance of the material. The nominal diffuse reflectance of the standard panel used is $\rho_d = 99$ %, yielding a theoretical BRDF value of $\rho_d/\pi = 0.31$ sr^{-1}.

Fig. 3(b) shows that the measured BRDF values at all four wavelengths are consistently concentrated around 0.31 sr^{-1}, in excellent agreement with the ideal Lambertian scattering behavior. Within the scattering angle range from −60° to 60°, the relative standard deviation is better than 1.27 %. These results demonstrate that the developed system exhibits high measurement accuracy and reliability.

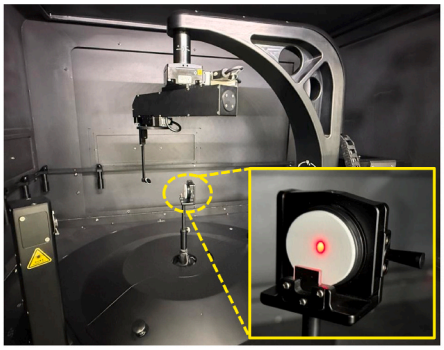
3. Sample surface characterization

In this study, the materials of gravitational wave telescope baffle using three different coating methods on the same material were investigated. The three samples are an uncoated aluminum alloy metal substrate, a Chemglaze Z306 coating, and a Black Anodize coating. The surface micro-morphology of the samples was measured using the LEXT^{TM} OLS5100 3D laser scanning microscope produced by OLYMPUS Corporation. The OLS5100 microscope utilizes a laser light source, focusing on the sample surface through the objective lens. The light that is reflected from the sample surface is directed through a confocal pinhole, ultimately reaching the detector. This confocal configuration allows the instrument to precisely measure surface morphology by selectively capturing light that originates solely from the focal plane. An objective lens magnification of 10x was selected for the measurement. The results are displayed in Fig. 4, with an image pixel size of 1024 × 1024 pixels and the actual measurement size being 1273 × 1280 μm.

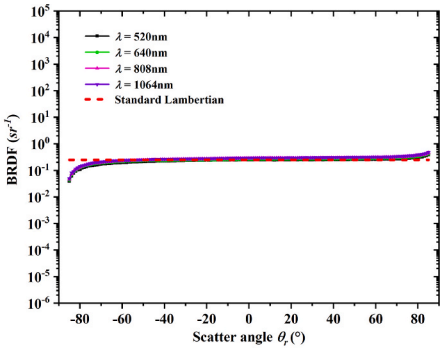
The surface of a material can be regarded as a collection of numerous randomly distributed microfacets. By analyzing the STL data of the surface's three-dimensional morphology using Python, the standard deviation (σ) of the microfacet slope can be statistically determined. This parameter is positively correlated with surface roughness and represents another way to quantify the microsurface morphology. A higher standard deviation of the slope indicates greater variation in the orientation of microfacets, leading to increased surface roughness. The surface roughness (Sa) and the standard deviation of the microfacet slope distribution (σ) are shown in Table 2.

4. Results and discussion

Utilizing the developed BRDF measurement apparatus, the BRDF of the metal substrate of the baffle and two types of matte black paint surfaces were measured. Considering the constantly changing incident angles of stray light inside the baffle, this paper conducted a detailed investigation of the BRDF of material surfaces at various incident angles. During the measurement, the incident angle range was set from 0° to 60° (with an interval of 10°). The scattering angle range was from −85° to 85° (with an interval of 1°). The test wavelengths covered 520 nm, 640

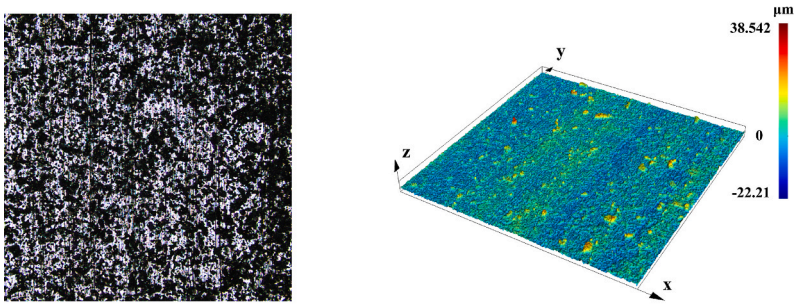


(a) System calibration using the Lambertian standard.

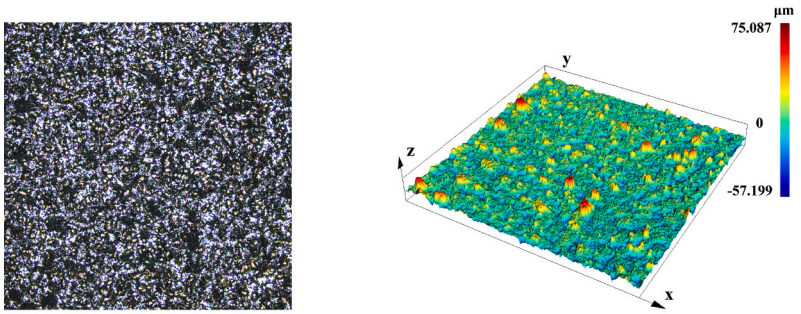


(a) BRDF curves at four wavelengths.

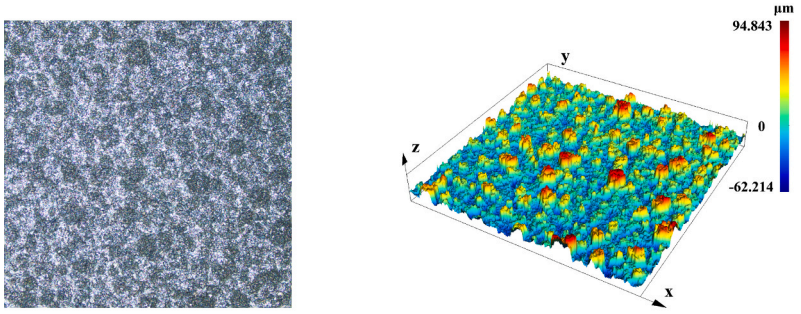
Fig. 3. BRDF of the Lambertian standard at four wavelengths.



(a) Baffle metal base (Aluminum alloy): 2D (left) 3D (right).



(b) Baffle black paint coating 01 (Chemglaze Z306): 2D (left) 3D (right).



(b) Baffle black paint coating 02 (Black Anodize): 2D (left) 3D (right).

Fig. 4. Surface micromorphology of three baffle samples (magnification: 10x).

Table 2
Gravitational wave telescope baffle surface parameters.

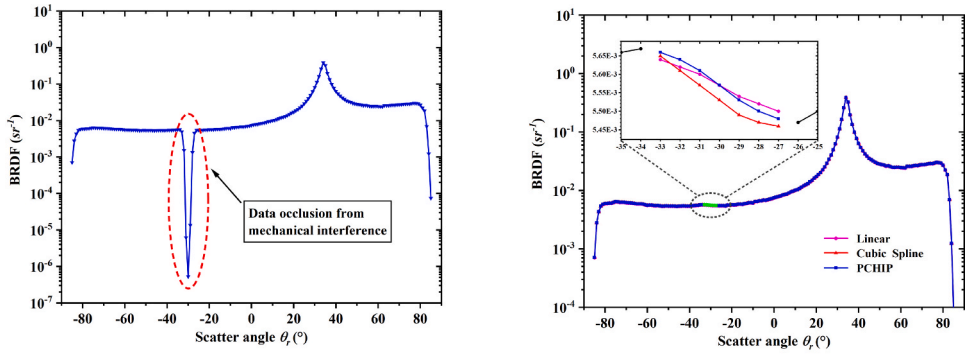
Type of material	Sa/μm	σ
Substrate (Aluminum alloy)	2.39	0.212
Coating 01 (Chemglaze Z306)	6.73	0.297
Coating 02 (Black Anodize)	14.41	0.347

nm, 808 nm, and 1064 nm. Due to geometric interference among the mechanical components of the experimental setup, when the scattering angle approaches the incident angle, the detector arm blocks the incident optical path over an angular range of approximately 6°, making measurements in this region unavailable, as shown in Fig. 5(a) (BRDF curve of the Aluminum alloy substrate, $\theta_i = 30°$, $\lambda = 1064$ nm). To fill this measurement blind region, linear interpolation, cubic spline interpolation, and piecewise cubic Hermite interpolating polynomial (PCHIP) methods are compared, as shown in Fig. 5(b). The results indicate that PCHIP interpolation preserves monotonicity and smoothness without introducing spurious oscillations; therefore, this method is ultimately adopted to reconstruct the missing data.

Fig. 6 presents the curves of the BRDF variation with incident angles for an aluminum alloy substrate under four wavelengths. It can be observed from the figure that, at all four wavelengths, the aluminum alloy substrate exhibits pronounced specular reflection characteristics. The BRDF increases with the scattering angle, reaching its peak when the scattering angle equals the incident angle, and then gradually decreases thereafter. At a wavelength of 640 nm (as shown in Fig. 6(b)), the BRDF exhibits a positive correlation with the incident angle, with the peak value of specular reflection increasing as the incident angle widens. Conversely, at a wavelength of 520 nm (as shown in Fig. 6(a)), an opposite distribution trend is observed: the BRDF value decreases from 0.203 sr^{-1} to 0.064 sr^{-1} as the incident angle increases from 0° to 50°, but unexpectedly rises to 0.092 sr^{-1} when the incident angle increases from 50° to 60°. This phenomenon might be related to the microstructure of the metal surface and its short-wave absorption characteristics.

Fig. 6(c) and (d) represent the BRDF curves for wavelengths of 808 nm and 1064 nm, respectively. It can be observed that at an incident angle of 30°, the peak value of specular reflection BRDF is the smallest compared to other incident angles. The BRDF peak values at each incident angle exhibit a "U" shaped distribution. Additionally, Fig. 6 reveals that the BRDF of the metal substrate includes a significant forward scattering component, a characteristic that becomes more pronounced at larger incident angles.

Fig. 7 shows the distribution curves of BRDF with changing incident angles for surfaces of Black Anodize and Chemglaze Z306 matte black paints. It is evident from the figure that, compared to the substrate, these



(a) BRDF data missing at certain detection angles. (a) Reconstruction of missing data.

Fig. 5. BRDF measurement blind region and comparison of interpolation reconstruction methods.

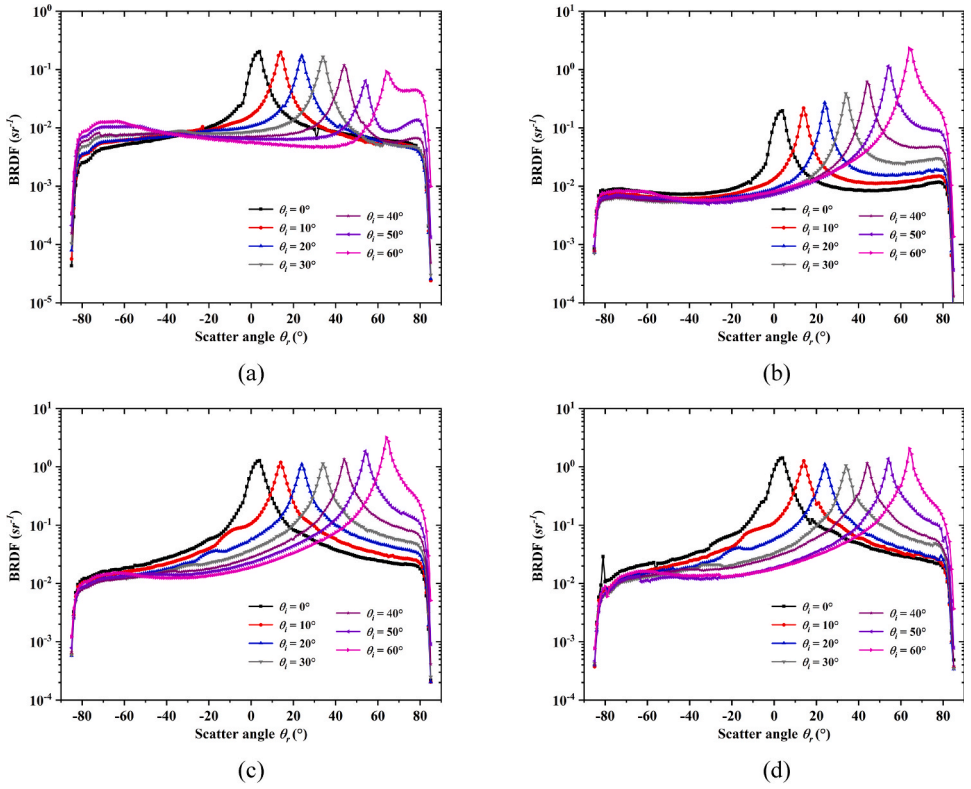


Fig. 6. BRDF measurements of the aluminum alloy substrate at four wavelengths with varying incident angles. (a)$\lambda = 520$ nm; (b)$\lambda = 640$ nm; (c) $\lambda = 808$ nm;(d) $\lambda = 1064$ nm.

two black paint materials demonstrate significantly stronger diffuse reflection characteristics. Comparing Figs. 7(b), (f), and 6(b), it is observed that at a wavelength of 640 nm, the BRDF for these two black paint coatings drops from 10^0 to around the 10^{-2} magnitude, achieving a two orders of magnitude attenuation of specular reflection. This indicates that both types of black paint coatings excel in suppressing specular reflection stray light. Fig. 7(e–h) indicate that when the laser incident angle is large, Black Anodize exhibits a specular reflection component, especially when the incident angle increases from 50° to 60°, the peak value of the BRDF significantly increases, reflecting an expansion in the angle at which reflection energy is concentrated. In contrast, the BRDF of Chemglaze Z306 shows a more stable trend with angle variation. At large incident angles, Black Anodize demonstrates significant forward scattering characteristics, with its BRDF specular reflection peak value being significantly higher than that of Chemglaze Z306. This indicates that Chemglaze Z306 is superior in suppressing grazing incidence stray light compared to Black Anodize.

At smaller incident angles, the BRDF value of Black Anodize is lower than that of Chemglaze Z306, a phenomenon closely related to surface roughness. Specifically, the surface roughness of Black Anodize is 14.41 μm, higher than that of Chemglaze Z306 at 6.73 μm. The increased roughness results in an enhanced masking effect, leading to greater absorption of beam energy, thereby reducing the material surface's reflective properties and enhancing its scattering characteristics. Additionally, the data further confirm that the background noise level measured by our constructed testing apparatus is superior to 1×10^{-5} sr^{-1}.

To more intuitively demonstrate the spectral BRDF characteristics of baffle material surfaces, the effects of four wavelengths: 520 nm, 640 nm, 808 nm, and 1064 nm, were studied under two incident angles of 20° and 60°, as shown in Fig. 8. The long-wave radiation characteristics of the three materials are evident, with the strongest BRDF surface

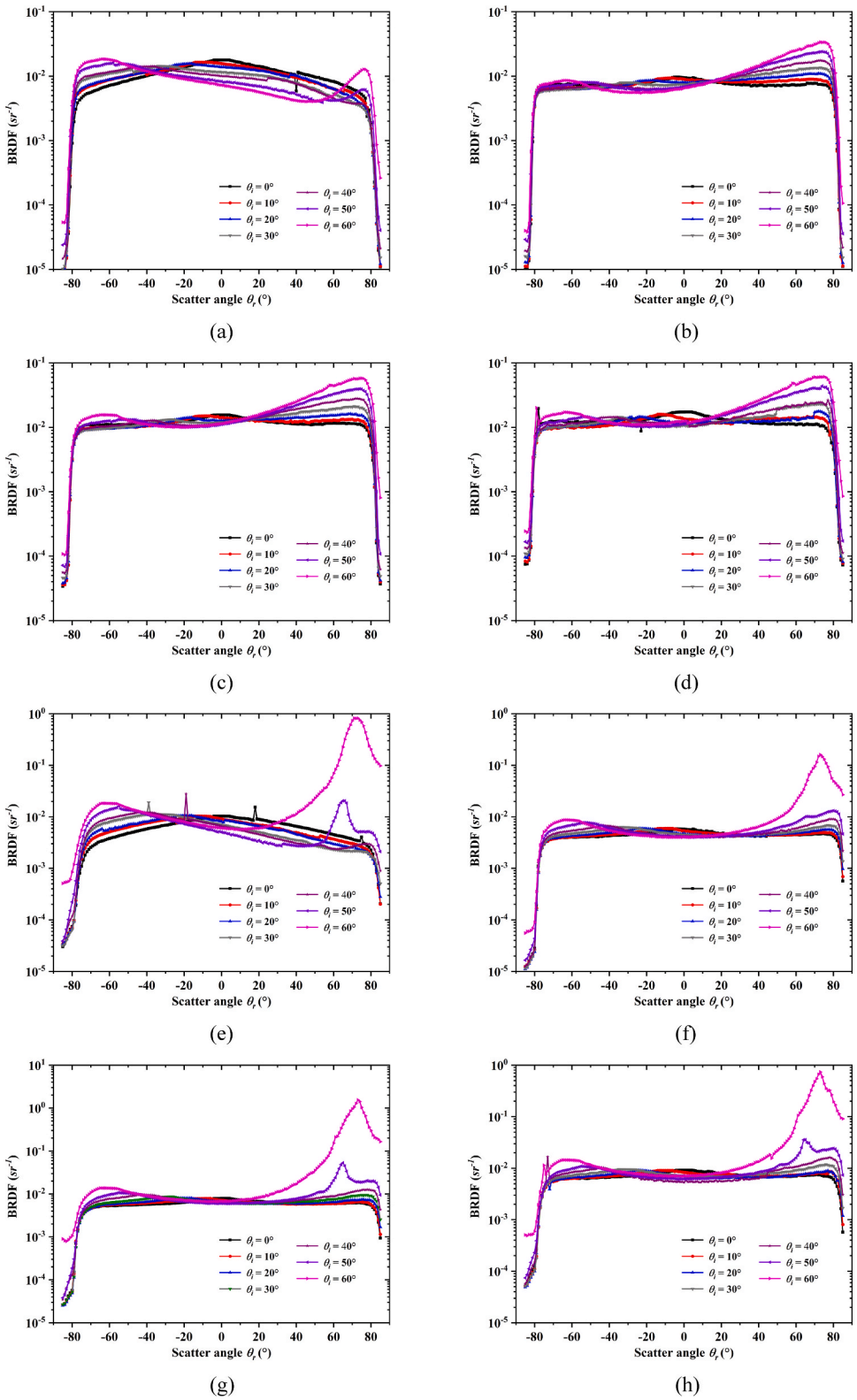


Fig. 7. BRDF of two black paint coatings at four wavelengths with varying incident angles. (a–d) Chemglaze Z306: (a)λ = 520 nm; (b)λ = 640 nm; (c) λ = 808 nm;(d) λ = 1064 nm; (e–h) Black Anodize: (e)λ = 520 nm; (f)λ = 640 nm; (g) λ = 808 nm;(h) λ = 1064 nm.

reflection characteristics at the near-infrared wavelength of 1064 nm, closely followed by the distribution at 808 nm. The BRDF shows a general increasing trend with increasing wavelength. From Fig. 8(a–b), it can be seen that the metal substrate has the weakest surface reflection characteristics at the short wavelength of 520 nm. With a larger incident angle of 60°, forward scattering is stronger, leading to a BRDF that first decreases and then increases, and the forward scattering component at 520 nm is greater than at the other three wavelengths.

Fig. 8(c–f) indicate that both Chemglaze Z306 and Black Anodize exhibit diffuse reflection effects at small incident angles, demonstrating

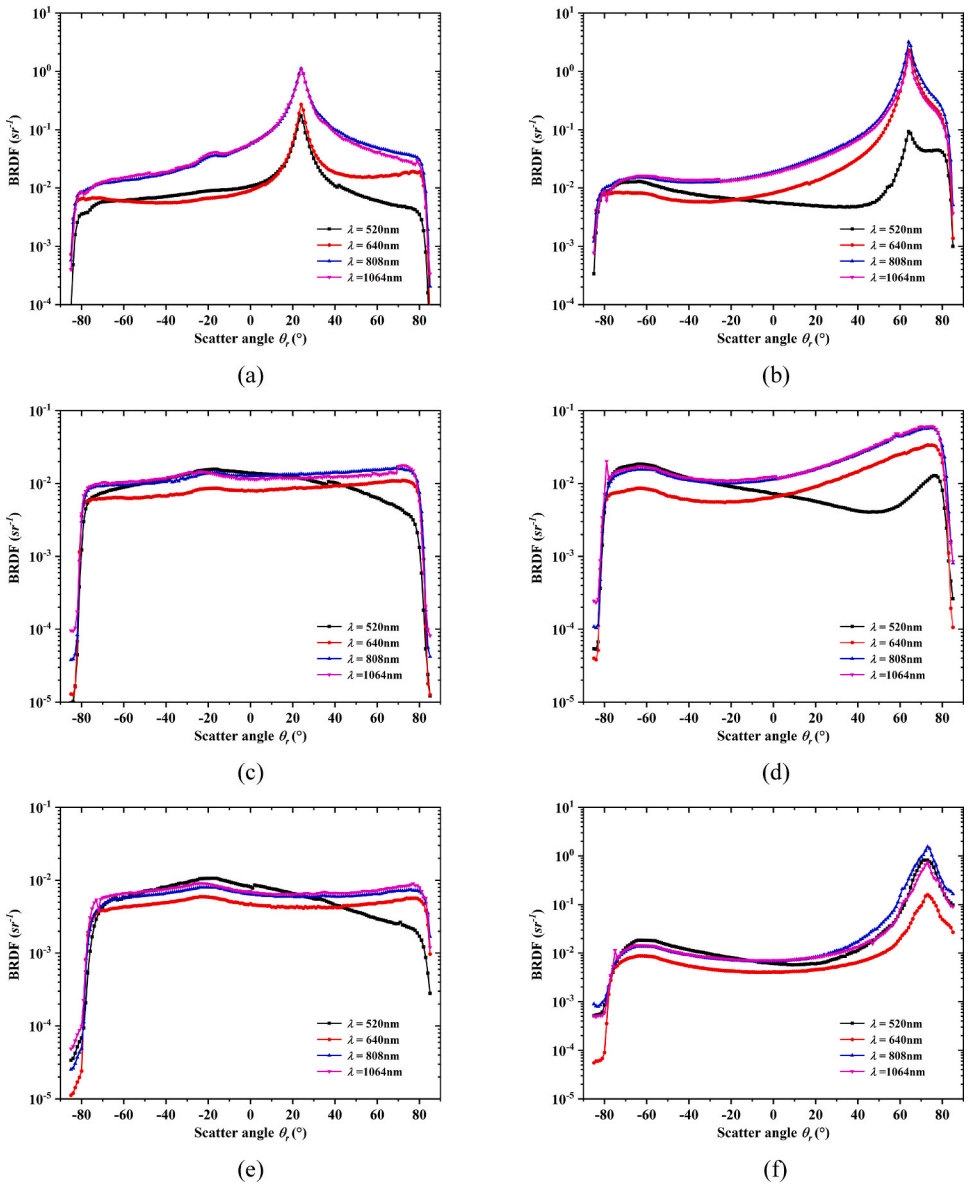


Fig. 8. BRDF of three materials at two incident angles under different laser wavelengths. (a–b) Aluminum alloy: (a) $\theta_i = 20°$; (b) $\theta_i = 60°$; (c–d) Chemglaze Z306: (c) $\theta_i = 20°$; (d) $\theta_i = 60°$; (e–f) Black Anodize: (e) $\theta_i = 20°$; (f) $\theta_i = 60°$.

that these materials effectively suppress specular components at four wavelengths and small angles. As seen in Fig. 8(f), Black Anodize exhibits specular reflection components at all four wavelengths, with the BRDF being minimal at normal incidence (near 0°) and slowly increasing from 0° to 60°, displaying an equal magnitude of forward scattering components that peak and then gradually decrease.

It is evident from these distribution patterns that, compared to the aluminum alloy metal substrate, the stray light suppression capability of Chemglaze Z306 and Black Anodize coatings is significantly enhanced by two orders of magnitude. At smaller incident angles, Black Anodize's suppression effect surpasses that of Z306, making it the preferred choice for applications requiring stray light reduction at smaller angles; for broader angle stray light suppression needs, Chemglaze Z306 demonstrates superior performance.

To further reveal the scattering characteristics of different baffle materials at the 1064 nm gravitational wave telescope's operating wavelength from the perspective of angular space structure, the angular distributions of the BRDF are represented in polar coordinates, as shown in Fig. 9. As can be seen from Fig. 9(a-b), the reflected energy of the aluminum alloy substrate remains highly concentrated in the specular reflection direction for all incident angles. Its polar coordinate distribution exhibits a distinctly narrow angle and strong directional characteristic, indicating a very high concentration of reflected energy within the angular space, easily forming high-intensity stray light paths in specific directions. In contrast, the Chemglaze Z306 and Black Anodize exhibit distinctly different angular spreading behaviors in their polar BRDF distributions.

As shown in Fig. 9(c–d), Chemglaze Z306 maintains a relatively stable and confined angular spread over the entire incident angle range of 0°-60°, without obvious directional transitions, thereby avoiding the re-concentration of reflected energy into a single angular region. By comparison, the Black Anodize shows forward angular compression of the scattering distribution under near-grazing incident conditions ($\theta_i =$ 60°), as illustrated in Fig. 9(e–f). Since the polar BRDF distributions intrinsically represent the angular allocation of reflected energy over the reflection hemisphere, quantitative evaluation of the overall reflective behavior of different materials under given incidence conditions can be achieved by integrating the BRDF over the reflection hemisphere. Under

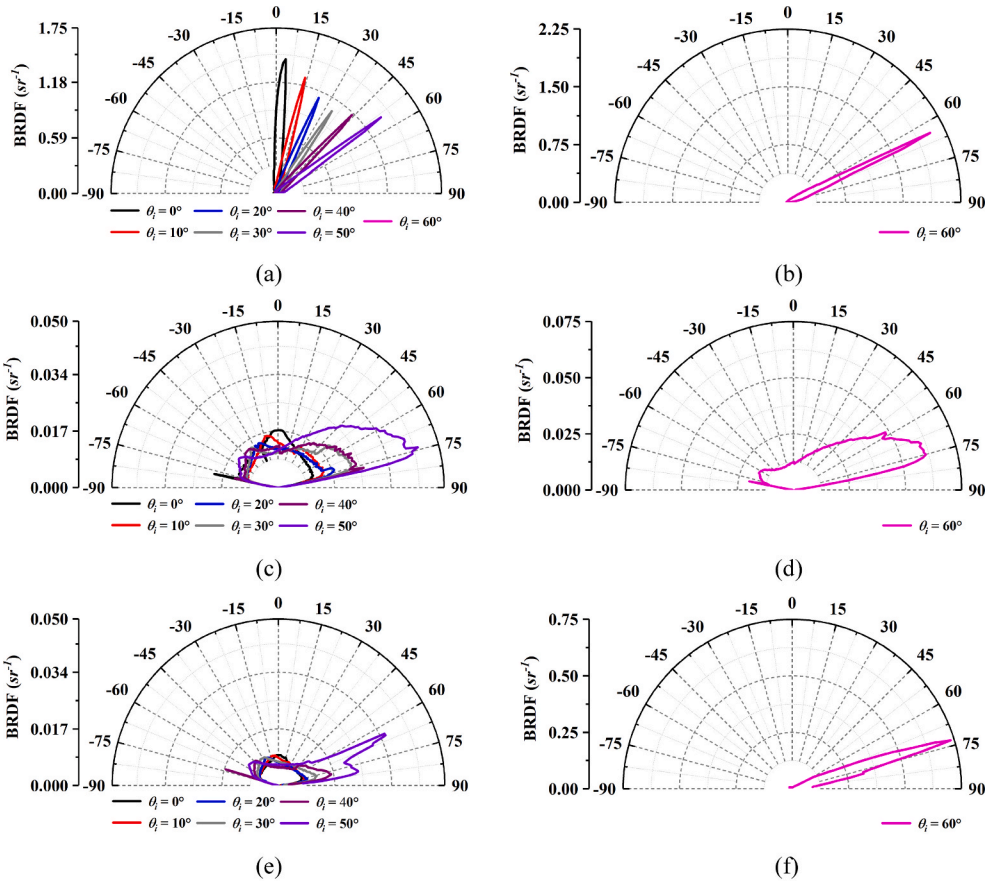


Fig. 9. Polar BRDF of three materials under different incident angles at 1064 nm. (a–b) Aluminum alloy: (a) $\theta_i = 0°$-50°; (b) $\theta_i = 60°$; (c–d) Chemglaze Z306: (c) $\theta_i = 0°$-50°; (d) $\theta_i = 60°$; (e–f) Black Anodize: (e) $\theta_i = 0°$-50°; (f) $\theta_i = 60°$.

unidirectional illumination, the relationship between the reflected radiance and the incident irradiance can be expressed as:

$$L_r(\theta_r, \varphi_r) = BRDF(\theta_i, \varphi_i, \theta_r, \varphi_r)E_i \tag{5}$$

The hemispherical reflected irradiance is given by:

$$E_r(\theta_i, \varphi_i) = \int_{\Omega^+} L_r(\theta_r, \varphi_r)\sin\theta_r\cos\theta_r d\theta_r d\varphi_r \tag{6}$$

By normalizing with respect to the incident irradiance, the directional-hemispherical reflectance (DHR) can be obtained as:

$$\rho_{\mathrm{DHR}}(\theta_i, \varphi_i) = \frac{E_r(\theta_i, \varphi_i)}{E_i} = \int_{\Omega^+} BRDF(\theta_i, \varphi_i, \theta_r, \varphi_r)\sin\theta_r\cos\theta_r d\theta_r d\varphi_r \tag{7}$$

Considering that the angular scattering of black coatings is statistically azimuthally isotropic, the above expression can be simplified as:

$$\rho_{\mathrm{DHR}}(\theta_i) = 2\pi \int_0^{\pi/2} BRDF(\theta_i, \varphi_i, \theta_r, \varphi_r)\cos\theta_r\sin\theta_r d\theta_r \tag{8}$$

Substituting the measured BRDF data into the above expressions, the calculated DHR values at an incident angle of 60° are 0.0702 for Chemglaze Z306 and 0.1485 for the Black Anodize. This indicates that, under the same oblique incidence conditions, the reflectance of Chemglaze Z306 is approximately half that of Black Anodize. This result further quantitatively verifies the stray light suppression advantage of Chemglaze Z306 under grazing incidence conditions.

5. BRDF simulation

5.1. BRDF model

BRDF is a complex function represented by multiple factors such as material surface reflectivity, roughness, surface topography distribution, and laser wavelength. Pure theoretical derivation methods will lead to problems such as low modeling accuracy. Therefore, the model parameters are inverted by actually measuring the BRDF on the surface of the structural parts to improve the accuracy of the model.

Building on the physical mechanisms of the interaction between light and surface morphology found in traditional studies, and considering the potential occlusion masking effect and multiple scattering phenomena on metallic surfaces, this paper divides the reflected energy from the material surface into three components: specular reflection, diffuse reflection, and multiple scattering, as shown in Eq. (9). The interaction between light and the material is illustrated in Fig. 10.

$$f_{BRDF}(\theta_i, \varphi_i, \theta_r, \varphi_r) = k_r \cdot f_r + k_d \cdot f_d + k_m \cdot f_m \tag{9}$$

in which, f_r, f_d and f_m respectively represent the specular reflection component, diffuse reflection component, and multiple scattering component. k_r, k_d and k_m represent the proportionality coefficients for each of these components, respectively.

The surface of the material can be regarded as composed of a large number of randomly distributed microfacets. Assuming α represents the angle between the normal of the microfacet and the normal of the macroscopic observed surface, and β is the incident angle relative to the microfacet normal. Then, based on geometric relations, the relationship between $\alpha, \beta, \theta_i, \theta_r, \varphi_i$, and φ_r satisfies the expression:

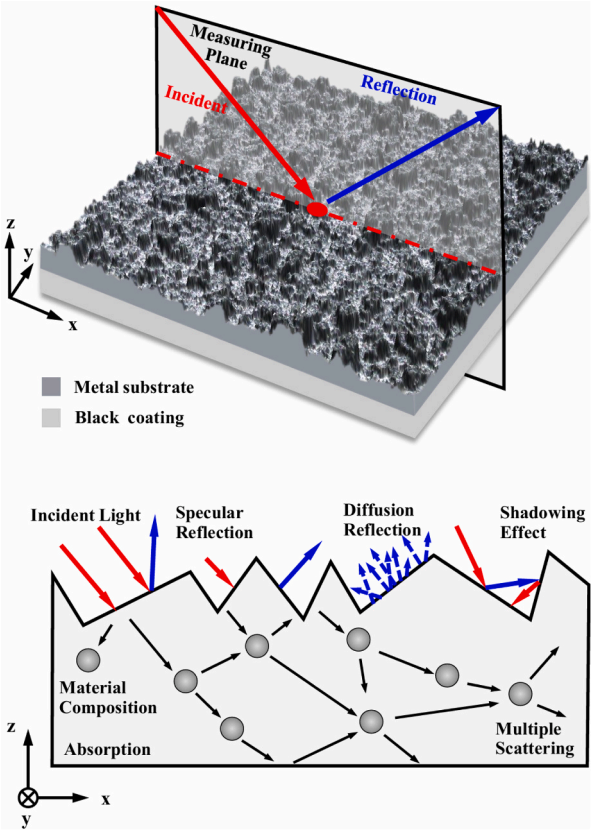


Fig. 10. Schematic representation of the interaction between light and material surface.

$$\begin{cases} \cos\alpha = \dfrac{\cos\theta_i + \cos\theta_r}{2\cos\beta} \\ \cos(2\beta) = \cos\theta_i\cos\theta_r + \sin\theta_i\sin\theta_r\cos(\varphi_r - \varphi_i) \end{cases} \tag{10}$$

The microfacet model typically employs the Torrance-Sparrow model [38] to describe the specular component of reflected light. The expression is as follows:

$$f_r(\theta_i, \varphi_i, \theta_r, \varphi_r) = \frac{P(\alpha)}{4\cos\alpha\cos\theta_i\cos\theta_r} G(\theta_i, \varphi_i, \theta_r, \varphi_r) F_f(\lambda) \tag{11}$$

wherein, $F_f(\lambda)$ is the Fresnel reflection coefficient, and $G(\theta_i, \varphi_i, \theta_r, \varphi_r)$ is the shadowing factor, which represents the shadowing and masking effect between adjacent facets on a rough surface. The expression is as follows:

$$G(\theta_i, \varphi_i, \theta_r, \varphi_r) = \min\left(1, \frac{2\cos\alpha\cos\theta_r}{\cos\beta}, \frac{2\cos\alpha\cos\theta_i}{\cos\beta}\right) \tag{12}$$

wherein, $P(\alpha)$ represents the distribution of microfacets. The Torrance-Sparrow model typically characterizes the distribution of microfacets on metal surfaces with a Gaussian distribution parameterized by the roughness constant. Research has shown that while the Gaussian distribution can accurately describe the microfacet distribution when the observation direction is near the specular reflection angle, its accuracy decreases when the observation direction deviates significantly from the specular reflection angle, especially in cases of high surface roughness. This lack of flexibility fails to adequately describe wide-angle non-specular reflections. To address this, the paper adopts a super-Cauchy distribution to characterize the distribution of microfacets on metal surfaces, as shown in the following equation:

$$P(\sigma, q) = \frac{(q-1)(\sqrt{2}\sigma)^{2q-2}}{\pi\cos^3\alpha\left[\tan^2\alpha + (\sqrt{2}\,\sigma)^2\right]^q} \tag{13}$$

in the formula, σ represents the standard deviation of the surface roughness of the material, the specific value of which has been calculated in Section 3. q is a constant related to the material properties. Therefore, the expression for the BRDF specular reflection component f_r is as follows:

$$f_r(\theta_i, \varphi_i, \theta_r, \varphi_r) = \frac{1}{4\pi\cos^4\alpha} \frac{(q-1)}{\cos\theta_i\cos\theta_r} \frac{(\sqrt{2}\sigma)^{2q-2}}{[\tan^2\alpha + (\sqrt{2}\sigma)^2]^q} G(\theta_i, \theta_r, \varphi) \tag{14}$$

This paper employs the Minnaert model to characterize diffuse reflection, which can effectively capture the influence of surface morphology on diffuse reflection properties. With just one additional parameter, the model significantly increases its adaptability to real-world scenarios. Due to its simplicity, the model is widely used in the field of space optics [39]. The expression is as follows:

$$f_d(\theta_i, \varphi_i, \theta_r, \varphi_r) = \frac{\rho_0}{\pi}(\cos\theta_i\cos\theta_r)^{k-1} \tag{15}$$

in which, k is a constant to be determined, with a value range of (0,1). When $k = 1$, the model can be transformed into the Lambert model.

Multiple scattering is phenomenologically similar to diffuse reflection. As surface roughness increases, the frequency of interactions between rough microfacets in the reflection direction increases upon departure from the surface, leading to a higher proportion of scattered energy and a reduced proportion of specular reflection energy. For the surface of baffle material, the spatial distribution of multiple reflections of light energy is uneven, and its magnitude is related to the reflection angle. According to the central limit theorem, its distribution tends to follow a standard normal distribution. Therefore, this paper represents the multiple scattering component in the following form:

$$f(\theta_i, \varphi_i, \theta_r, \varphi_r) = \frac{1}{\sqrt{2\pi}\sigma_m}\exp\left(-\theta_r^2/2\sigma_m^2\right) \tag{16}$$

In the above equation, σ_m is a parameter to be determined. By combining Eqs. (14), (15), and (16), the BRDF model for the surface of the sample material can be derived as follows:

$$f_{BRDF}(\theta_i, \varphi_i, \theta_r, \varphi_r) = k_r \frac{1}{4\pi\cos^4\alpha} \frac{(q-1)}{\cos\theta_i\cos\theta_r} \frac{(\sqrt{2}\sigma)^{2q-2}}{\left[\tan^2\alpha + (\sqrt{2}\sigma)^2\right]^q} G(\theta_i, \varphi_i, \theta_r, \varphi_r) + k_d \frac{1}{\pi}(\cos\theta_i\cos\theta_r)^{k-1} + k_m \frac{1}{\sqrt{2\pi}\sigma_m}\exp\left(-\theta_r^2/2\sigma_m^2\right) \tag{17}$$

5.2. Data fitting

The Simulated Annealing (SA) algorithm solves combinatorial optimization problems by mimicking the physical process of annealing to find the optimal solution [40]. It is known for its simplicity in computation and strong robustness. The minimum root mean square error between the model calculations and the measured data is used as the objective function to determine the optimal parameters of the model. The objective function can be expressed as:

$$E(k_r, k_d, k_m, q, k, \sigma_m) = \min \sum_{\theta_i}\sum_{\theta_r} g_1(\theta_i) g_2(\theta_r) \times \frac{\left[f_{\text{model}}\ (k_r, k_d, k_m, q, k, \sigma_m) - f_{\text{measured}}\ (\theta_i, \theta_r)\right]^2}{\sum_{\theta_i}\sum_{\theta_r} g_1(\theta_i) g_2(\theta_r) \left[f_{\text{measured}}\ (\theta_i, \theta_r)\right]^2} \tag{18}$$

In the equation, f_{model} represents the model's calculated value, while

Table 3
Model fitting parameters and fitting errors of three baffle materials.

Material	Sa/μm	λ/nm	k_r	k_d	k_m	q	k	σ_m	δ_{model}	δ_{max}
Aluminum alloy	2.39	520	68.33	2.15	−33.67	2.89	0.20	0.98	7.03%	7.36%
		640	47.17	2.46	−40.10	1.96	0.08	0.85	5.68%	6.08%
		808	52.89	2.95	−42.89	2.38	0.21	0.98	8.67%	8.94%
		1064	58.85	3.13	−28.51	2.18	0.08	0.79	6.35%	6.81%
Chemglaze Z306	6.73	520	16.89	20.81	3.62	1.11	0.08	0.68	4.23%	4.76%
		640	18.84	18.97	2.94	2.02	0.21	0.71	5.69%	6.09%
		808	17.69	16.39	2.47	1.37	0.08	0.80	4.96%	5.42%
		1064	18.82	23.47	2.02	1.29	0.13	0.31	3.92%	4.86%
Black Anodize	14.41	520	11.84	22.86	6.85	1.98	0.76	0.68	5.75%	6.15%
		640	10.96	26.51	7.65	1.11	0.68	0.99	4.66%	5.14%
		808	10.46	19.68	8.13	2.94	0.83	0.81	6.25%	6.62%
		1064	8.65	17.46	5.31	1.71	0.78	0.73	5.61%	6.02%

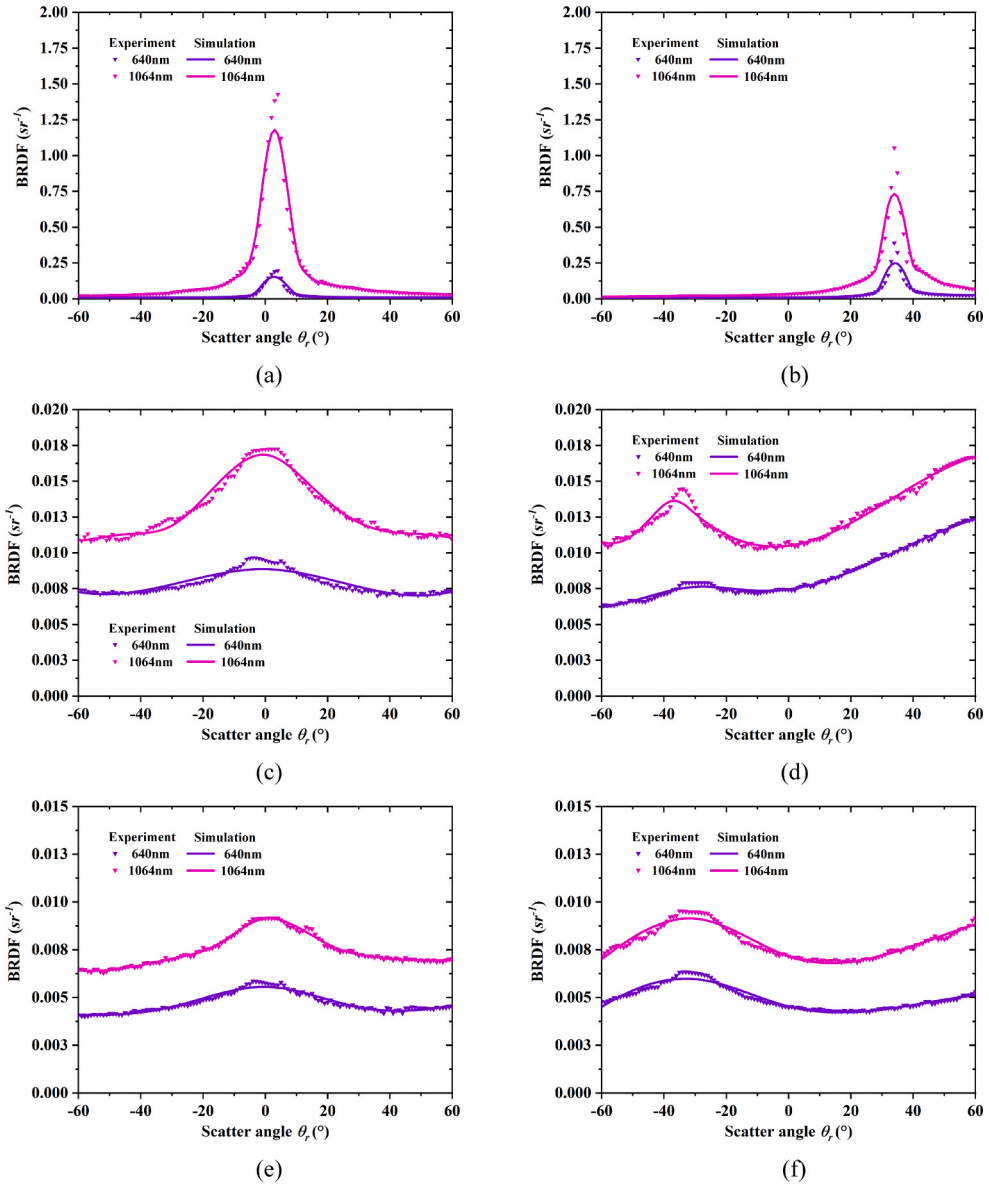


Fig. 11. Comparison between measured data and simulation results for two incident angles at 640 nm and 1064 nm. (a-b) Aluminum alloy: (a) $\theta_i = 0°$; (b) $\theta_i = 30°$; (c–d) Chemglaze Z306: (c) $\theta_i = 0°$ (d) $\theta_i = 30°$; (e–f) Black Anodize: (e) $\theta_i = 0°$; (f) $\theta_i = 30°$.

f_{measured} is the measured value; $g_1(\theta_i)$ and $g_2(\theta_r)$ are weighting functions that adjust the contribution of each error to the total error. The Relative Root Mean Square Error (RMSE) quantifies the relative deviation between simulation and measurement data and is expressed as:

$$\delta_{model} = \sqrt{\frac{1}{N}\sum\left(\frac{f_{\text{model}} - f_{\text{measured}}}{f_{\text{measured}}}\right)^2} \tag{19}$$

However, δ_{model} does not account for experimental measurement uncertainty, as previously listed in Table 1. To address this, the error propagation method is applied to incorporate experimental measurement uncertainty into the overall error calculation, enhancing model accuracy. Specifically, the combined error δ_{max} is obtained by integrating the model prediction error δ_{model} with the measurement uncertainty (BRDF)/BRDF for each material:

$$\delta_{\max} = \sqrt{\delta_{\text{model}}^2 + (U(BRDF)/BRDF)^2} \tag{20}$$

The calculated model fitting parameters and associated fitting errors are summarized in Table 3. The results show strong agreement between the model predictions and the experimental measurements, although minor discrepancies remain. The maximum fitting error is 8.94%. Fig. 11 compares the BRDF modeling data and measured values for three baffle samples at two representative incident angles of 0° and 30°, with wavelengths of 640 nm and 1064 nm. It can be seen that the proposed model successfully reproduces the main scattering characteristics observed experimentally, including the specular reflection peak at small scattering angles and the overall angular dependence of the diffuse scattering component. These results confirm that the developed BRDF model can accurately and comprehensively characterize the BRDF scattering properties of materials used in space gravitational wave telescope baffles.

6. Conclusion

In this study, an automated BRDF measurement system was constructed, achieving a measurement uncertainty of 2.18%. The BRDF patterns of an aluminum alloy metal substrate and two types of matte black paint coatings for gravitational wave telescope baffle were measured and analyzed, exploring the effects of incident angles, surface micro-topography, and wavelength on BRDF. A BRDF model for the surface of the baffle materials was established, and the simulated annealing algorithm was used to fit the measured data and obtain the optimal parameters. The results indicate the following main conclusions:

(1) The BRDF characteristics of the baffle material surface are closely related to the angle and wavelength of the incident light, with the surface reflection properties intensifying as the wavelength increases. Under conditions of short wavelength and large incident angle, the phenomenon of forward scattering is significantly enhanced. Particularly, at a large incident angle of 60°, the Black Anodize coating exhibits equal amounts of forward scattering across all four wavelengths.
(2) The BRDF of the aluminum alloy substrate exhibits significant specular reflection characteristics. Compared to the aluminum alloy substrate, the Black Anodize and Chemglaze Z306 black paint coatings achieve an attenuation of specular reflection by about two orders of magnitude, demonstrating excellent stray light suppression capabilities. Especially at smaller incident angles, the suppression effect of Black Anodize exceeds that of Chemglaze Z306, while Chemglaze Z306 performs better in suppressing grazing incident stray light.
(3) The established BRDF model integrates specular, diffuse, and multiple scattering effects, and incorporates a super-Cauchy distribution to refine the description of the material surface structure, effectively characterizing the relationship between micromorphology and reflective properties. The model's fitting error with actual measurements is better than 8.94%, confirming the model's feasibility. The constructed BRDF model can serve as an input to optical simulation platforms, enabling stray light path tracing and energy distribution analysis, thereby supporting accurate performance evaluation of space gravitational wave detection instruments.

CRediT authorship contribution statement

Bolun Ma: Writing – original draft, Visualization, Methodology. **Qinfang Chen:** Supervision, Resources, Formal analysis. **Hu Wang:** Supervision, Resources, Funding acquisition. **Yaoke Xue:** Validation, Software, Project administration. **Zhanpeng Ma:** Project administration, Investigation. **Jiawen Liu:** Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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